

Table 1. Prediction results on microbiome datasets

Continuous and binary outcomes

Continuous Outcomes							
Data Source	Outcome	Data Size	Est. S–N Ratio	treeFA Loss	Rare Loss	Lasso Loss	Ridge Loss
Sinha et al. (2016)	BMI	131 × 85	–	19.908	20.042	37.348	114.617
Erawijantari et al. (2020)	BMI	96 × 1921	–	11.157	11.267	90.362	144.612
Wang et al. (2020)	BMI	280 × 2853	–	17.901	18.386	45.554	451.384
Yachida et al. (2019)	BMI	347 × 3418	–	10.954	10.586	1696.004	173.833
Binary Outcomes							
Data Source	Outcome	Data Size	treeFA AUC / Rare AUC	treeFA Loss	Rare Loss	Lasso Loss	Ridge Loss
Sinha et al. (2016)	CRC	131 × 85	0.572/0.574	0.638	0.633	0.625	0.635
Kim et al. (2020)	CRC	240 × 121	0.514/0.521	0.690	0.692	0.698	0.690
Jacobs et al. (2016)	IBD	90 × 507	0.647/0.582	0.667	0.704	0.625	0.725
Erawijantari et al. (2020)	GC*	96 × 1921	0.543/0.542	0.701	0.698	0.726	0.730
Franzosa et al. (2019)	IBD*	220 × 2834	0.833/0.823	0.435	0.451	0.474	0.553
Wang et al. (2020)	ESRD*	287 × 2853	0.835/0.831	0.422	0.420	0.376	0.687
Yachida et al. (2019)	CRC	347 × 3418	0.533/0.523	0.661	0.661	0.687	0.671